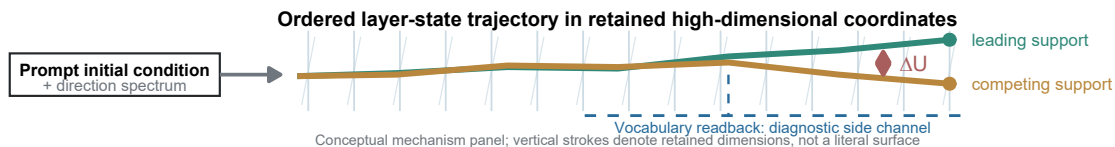


High-dimensional layer-state trajectories organize the retained observables

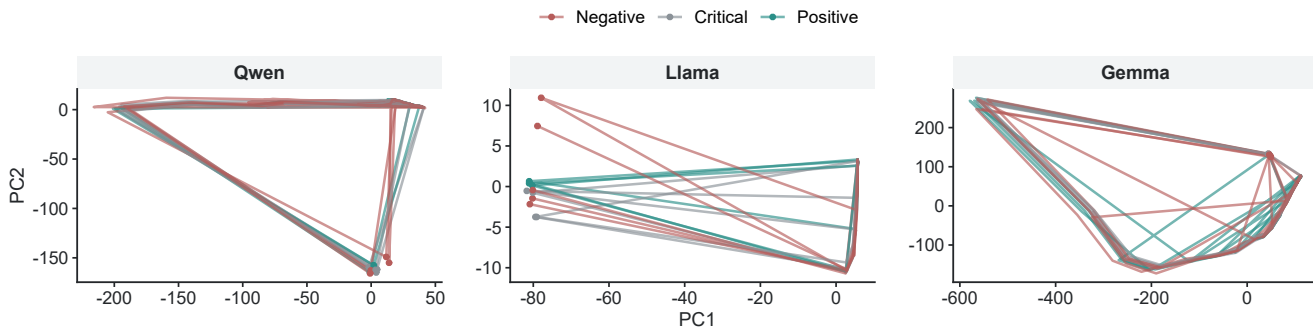
a



b

Low-dimensional orientation aid from condition-centroid trajectories

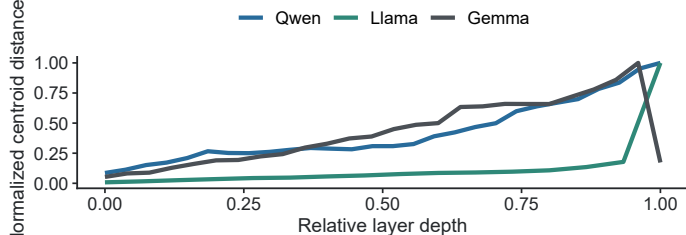
PCA is fitted separately within each checkpoint; axes are not mechanism or cross-model coordinates



c

Condition separation evolves across depth

High-dimensional centroid distance; normalized within checkpoint



d

Visualization boundary
Panels b-d are projections or summaries
of high-dimensional hidden states
No literal 3D manifold is claimed

e

High-dimensional barcode retains structure beyond a 3D projection

Range across condition centroids in the 32 highest-variance hidden dimensions; square-root colour scale

